

FBISE Class 9 Mathematics
Comprehensive Test
Chapter 5: Linear Equations and Inequalities
Prepared by Fakhar STEM Sphere

Name: _____ **Roll No.:** _____ **Total Marks: 80**

Class: 9 Subject: Mathematics Time: _____

Instructions: Section A contains 20 MCQs. Section B contains 15 short questions. Each short question carries 4 marks. Solve all questions neatly.

Section A - Objective Type

20 Marks

Q.1 Choose the correct option.

1. $-2x \leq 4$ gives:

- (A) $x \leq -2$ (B) $x \geq -2$ (C) $x \leq -1$ (D) $x = -1$

2. $x + 3 > 5$ implies:

- (A) $x = 0$ (B) $x < 2$ (C) $x \leq 2$ (D) $x > 2$

3. Solution of the inequality $-2x - \frac{1}{2} \leq \frac{3}{2}$ is:

- (A) $x > -1$ (B) $x < -1$ (C) $x \geq -1$ (D) $x \leq -1$

4. A pizza costs Rs. 1350, and you have Rs. 2500. The maximum amount of pizza that can be bought is represented by:

- (A) $x \geq 1.85$ (B) $x \leq 1.85$ (C) $x > 1.85$ (D) $x = 2$

5. The solution to the inequality $x + 3 < 5$ is:

- (A) $x > 2$ (B) $x < 2$ (C) $x = 2$ (D) $x \geq 2$

6. If x is smaller than 10, then which statement is true?

- (A) $x \geq 20$ (B) $x \leq 10$ (C) $x < 10$ (D) $x > 10$

7. Which of the following is a solution of the inequality $3 - 4x \leq 11$?

- (A) -8 (B) -2 (C) $-\frac{14}{4}$ (D) None of these

8. $x = \underline{\hspace{1cm}}$ is a solution of the inequality $-2 < x < \frac{3}{2}$.

- (A) -5 (B) 3 (C) 0 (D) $\frac{3}{2}$

9. If x is no larger than 10, then:

- (A) $x \geq 8$ (B) $x \leq 10$ (C) $x < 10$ (D) $x > 10$

10. If the capacity c of an elevator is at most 1600 pounds, then:

- (A) $c < 1600$ (B) $c \geq 1600$ (C) $c \leq 1600$ (D) $c > 1600$

11. $x = 0$ is a solution of the inequality:
 (A) $x > 0$ (B) $3x + 5 < 0$ (C) $x + 2 < 0$ (D) $x - 2 < 0$
12. Which one is the standard form of a linear equation?
 (A) $ax^2 + b = 0$ (B) $ax + b = -c$ (C) $ax + b > 0$ (D) $ax + b = 0$
13. What number must be subtracted from the right side of $7x = 30$ so that 4 is a solution of the resulting equation?
 (A) 7 (B) 2 (C) 4 (D) -2
14. Which one is the solution of $\frac{4}{x} - \frac{2}{x} = 5$?
 (A) -1 (B) $\frac{2}{5}$ (C) $\frac{5}{2}$ (D) zero
15. Which one is a strict inequality?
 (A) $x + 3 \neq 0$ (B) $12x > 5$ (C) $2y - 3 \leq 0$ (D) $4x + 5 \geq 0$
16. Which one is the solution of $2(x + 6) \leq 3(x + 4)$?
 (A) $x \leq 0$ (B) $x \geq 0$ (C) $x \leq 24$ (D) $x \geq 24$
17. Which is the solution set of $|x| + 7 = 3$?
 (A) $\{-4\}$ (B) $\{4, -4\}$ (C) $\{\}$ (D) $\{-7, -3\}$
18. Which one is the solution set of $\sqrt{5x} = -10$?
 (A) $\{\}$ (B) $\{-20\}$ (C) $\{20\}$ (D) $\{-2\}$
19. Which one is the solution set of $5|x| = 25$?
 (A) $\{-25, 25\}$ (B) $\{-25\}$ (C) $\{5\}$ (D) $\{-5, 5\}$
20. Which one is the solution set of $\sqrt{3x + 1} = 5$?
 (A) 25 (B) 8 (C) 24 (D) $\frac{26}{3}$

Section B - Short Questions

60 Marks

Q.2 Solve the following questions. Each question carries 4 marks.

- (i) Solve the inequality and plot the solution on the real number line: **[4 Marks]**

$$-3x + 11 \geq 5(x + 7) - 32 \geq 3x - 1, \quad x \in \mathbb{R}$$

- (ii) Solve the following inequalities: **[4 Marks]**

$$2x - 3 < x + 1 \quad \text{and} \quad 3x - 3 \geq 2x + 8$$

(iii) Solve the inequality and plot the solution on the real number line: **[4 Marks]**

$$3x + 5 \leq 5 - 3(x + 2) \leq 6x - 10, \quad x \in \mathbb{R}$$

(iv) Solve the linear equation: **[4 Marks]**

$$\frac{1}{3}(x - 2) + \frac{2 - 3x}{2} = \frac{x + 5}{6}$$

(v) Solve the given linear equation and represent the solution on a real number line: **[4 Marks]**

$$\frac{5}{6}x - \frac{7}{3} = \frac{3}{4}x + \frac{5}{2}$$

(vi) Solve the equation and represent the solution on a real number line: **[4 Marks]**

$$\frac{3x - 5}{4} = \frac{2x + 1}{3}$$

(vii) Solve: **[4 Marks]**

$$\frac{3x}{5} - \frac{1}{2} = \frac{x}{4} + 1$$

(viii) Solve: **[4 Marks]**

$$\frac{2x + 3}{5} = \frac{3 - 4x}{8}$$

(ix) Solve: **[4 Marks]**

$$\frac{4(x + 2)}{3} - \frac{6(x - 7)}{7} = 12$$

(x) Solve: **[4 Marks]**

$$\frac{1}{5}(x - 8) + \frac{4 + x}{7} = 7 - \frac{23 - x}{5}$$

(xi) Solve: **[4 Marks]**

$$4 + 2\sqrt{3y + 1} = 3$$

(xii) Solve: **[4 Marks]**

$$\sqrt{\frac{a+6}{a+2}} = \sqrt{\frac{a+2}{a-1}}$$

(xiii) Solve: **[4 Marks]**

$$\sqrt{9-2x} = \sqrt{5x-12}$$

(xiv) Solve, where $x \in \mathbb{R}$: **[4 Marks]**

$$|2x+5| = 11$$

(xv) Solve the absolute value equation, where $x \in \mathbb{R}$: **[4 Marks]**

$$\frac{|10-x|}{5} = \frac{|2x-5|}{2}$$